Week 5

Systems of linear equations (Section 3.1)

$$\begin{array}{rcl} D &=& 2S \\ D &=& C+3 \\ D+S+C &=& 17 \end{array} \qquad\qquad \begin{array}{rcl} x_1 - 2x_2 &=& 0 \\ x_1 - x_3 &=& 3 \\ x_1 + x_2 + x_3 &=& 17 \end{array}$$

children's age puzzle (Section 0.3) standard form ($x_1 = D, x_2 = S, x_3 = C$)

Definition 3.1: A *system of linear equations* in m equations and n variables $x_1, x_2, \ldots, x_n$ is of the form

$$\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &=& b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &=& b_2 \\ &\vdots& \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &=& b_m \end{array}$$

The a_{ij} and b_i: **known** real numbers The x_j: **unknown** real numbers (to be computed)
Matrix-vector form:

Puzzle:

$$A\mathbf{x} = \mathbf{b}: \quad \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}}_{A,\ m\times n} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\mathbf{x}\in\mathbb{R}^n} = \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\mathbf{b}\in\mathbb{R}^m} \qquad \underbrace{\begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}}_{A} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 0 \\ 3 \\ 17 \end{pmatrix}}_{\mathbf{b}}$$

A: *coefficient matrix* $\mathbf{x}$: *vector of variables* $\mathbf{b}$: *right-hand side*
Solving the system: compute $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$ (or report that there is no such vector)!

A in row notation

$$A = \begin{bmatrix} — & \mathbf{u}_1^\top & — \\ — & \mathbf{u}_2^\top & — \\ & \vdots & \\ — & \mathbf{u}_m^\top & — \end{bmatrix}:$$

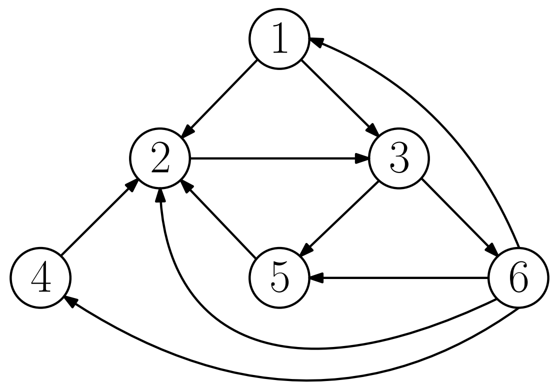
Simultanously fulfill the m equations $\mathbf{u}_i^\top \mathbf{x} = b_i, \quad i = 1, 2, \ldots m$!

A in column notation

$$A = \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix}:$$

Write $\mathbf{b}$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$: $\mathbf{b} = \sum_{j=1}^{n} x_j \mathbf{v}_j$!

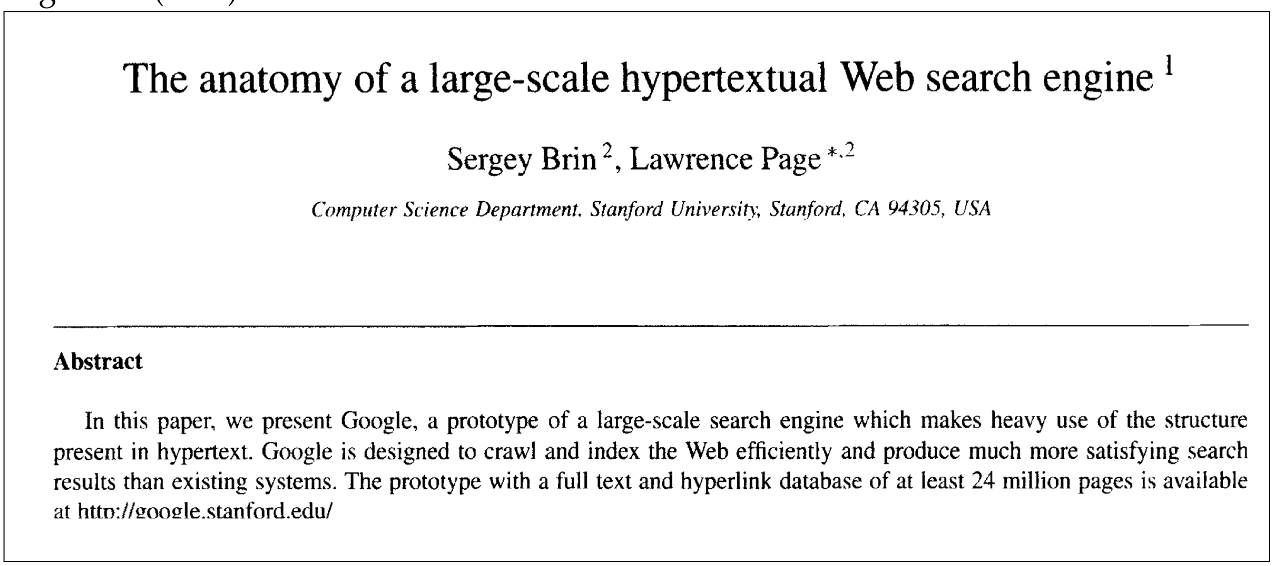
The PageRank algorithm: works on *link graph* (circles: web pages, arrows: links)



Which page is most relevant?

Old school measure: number of *citations* (links to the page): page 2 (4 citations) wins.

PageRank (1998):



The anatomy of a large-scale hypertextual Web search engine [1]

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Abstract

In this paper, we present Google, a prototype of a large-scale search engine which makes heavy use of the structure present in hypertext. Google is designed to crawl and index the Web efficiently and produce much more satisfying search results than existing systems. The prototype with a full text and hyperlink database of at least 24 million pages is available at http://google.stanford.edu/

Principles:

- Citation from a relevant page counts more.
- Citation from a page that cites many pages counts less.

$\rightarrow$ relevance: sum of relevances of citing pages, divided by number of pages cited:

$$x_2 = \frac{x_1}{2} + x_4 + x_5 + \frac{x_6}{4} \quad \text{(same for the other 5 pages)}$$

System of 6 linear equations in 6 variables! But with useless solution $\mathbf{0}$.

Fix: use *damping factor* d close to 1 (for example, $d = 7/8$):

$$x_2 = (1-d) + d\left(\frac{x_1}{2} + x_4 + x_5 + \frac{x_6}{4}\right)$$

Unique solution (rounded): page 3 (rank 1.7307) wins.

$$x_1 = 0.31797, x_2 = 1.6761, x_3 = 1.7307, x_4 = 0.31797, x_5 = 1.0751, x_6 = 0.88217.$$

Matrix inversion: An $m \times m$ matrix A is invertible $\Leftrightarrow$ there exsist an $m \times m$ matrix B (the inverse) with $AB = I$ (Def. 2.57).

$$B = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_m \\ | & | & & | \end{bmatrix} : \qquad \underbrace{\begin{bmatrix} | & | & & | \\ A\mathbf{x}_1 & A\mathbf{x}_2 & \cdots & A\mathbf{x}_m \\ | & | & & | \end{bmatrix}}_{AB} = \underbrace{\begin{bmatrix} | & | & & | \\ \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_m \\ | & | & & | \end{bmatrix}}_{I}$$

To find $\mathbf{x}_j$, solve $A\mathbf{x} = \mathbf{e}_j$! To find $B = A^{-1}$, solve m systems of linear equations!

Gauss elimination (Section 3.2)

Algorithm for solving $A\mathbf{x} = \mathbf{b}$ with square matrix ($m \times m$):

1. Transform$A\mathbf{x} = \mathbf{b}$ into a system $U\mathbf{x} = \mathbf{c}$ with the same solutions but "nice" matrix U (upper-triangular)
2. $U\mathbf{x} = \mathbf{c}$ can be solved easily.

Back substitution: if U is upper triangular

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 19 \\ 17 \\ 14 \end{pmatrix} :$$

equation	before substitution	after substitution	solution
1	$2x_1 + 3x_2 + 4x_3 = 19$	$2x_1 + 11 = 19$	$x_1 = 4$
2	$5x_2 + 6x_3 = 17$	$5x_2 + 12 = 17$	$x_2 = 1$
3	$7x_3 = 14$		$x_3 = 2$

Case $m \times m$: Equation $i = m, m-1, \ldots, 1$:

$$\sum_{j=i}^{m} u_{ij} x_j = c_i, \text{ or (when solved for } x_i\text{):} \quad x_i = \frac{c_i - \sum_{j=i+1}^{m} u_{ij} x_j}{u_{ii}}.$$

Already know $x_m, x_{m-1} \ldots, x_{i+1}$, now we also know x_i! Needs $u_{ii} \neq 0$!

Algorithm 1:

function BACK SUBSTITUTION($U, \mathbf{c}$) $\triangleright\, U \in \mathbb{R}^{m \times m}, \mathbf{c} \in \mathbb{R}^m$

  $\mathbf{x} \leftarrow \mathbf{0} \in \mathbb{R}^m$

  for $i = m, m-1, \ldots, 1$ **do**

    $x_i \leftarrow \dfrac{c_i - \sum_{j=i+1}^{m} u_{ij} x_j}{u_{ii}}$ $\triangleright \leftarrow$: assignment to a (programming language) *variable*

  end for

  return $\mathbf{x}$

end function

Elimination: If A not upper triangular
$A\mathbf{x} = \mathbf{b} \to U\mathbf{x} = \mathbf{c}$ (same solutions, U upper triangular)

$$\underbrace{\begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix}}_{A} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 19 \\ 55 \\ 50 \end{pmatrix}}_{\mathbf{b}}$$

Get rid of 4: subtract $2 \cdot$ (equation 1) from equation 2:

$$\begin{array}{rrcrcrcrcr} & \text{(equation 2)} & : & 4x_1 & + & 11x_2 & + & 14x_3 & = & 55 \\ - & 2 \cdot \text{(equation 1)} & : & 4x_1 & + & 6x_2 & + & 8x_3 & = & 38 \\ \hline & \text{(equation 2')} & : & & & 5x_2 & + & 6x_3 & = & 17 \end{array}$$

$\to A'\mathbf{x} = \mathbf{b}'$:

$$\underbrace{\begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix}}_{A'} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 19 \\ 17 \\ 50 \end{pmatrix}}_{\mathbf{b}'}.$$

This was a *row subtraction*: matrix transformation applied to all columns of A and $\mathbf{b}$:

$$T_{E_{21}} : \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \mapsto \begin{pmatrix} v_1 \\ v_2 - 2v_1 \\ v_3 \end{pmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

E_{21}: *elimination matrix*
$A' = E_{21}A, \quad \mathbf{b}' = E_{21}\mathbf{b}$ — E_{21}: subtract $2\cdot$(row 1) from (row 2)
Can undo it:
$A = E_{21}^{-1}A', \quad \mathbf{b} = E_{21}^{-1}\mathbf{b}'$ — E_{21}^{-1}: add $2\cdot$(row 1) to (row 2)

$A\mathbf{x} = \mathbf{b}$ and $A'\mathbf{x} = \mathbf{b}'$ have the same solutions:

- If $A\mathbf{x} = \mathbf{b}$, then $A'\mathbf{x} = E_{21}A\mathbf{x} = E_{21}\mathbf{b} = \mathbf{b}'$
- If $A'\mathbf{x} = \mathbf{b}'$, then $A\mathbf{x} = E_{21}^{-1}A'\mathbf{x} = E_{21}^{-1}\mathbf{b}' = \mathbf{b}$

Column by column, eliminate all red entries ($\to$ upper triangular system from before):

fat number: the **pivot**

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \qquad \mathbf{b} = \begin{pmatrix} 19 \\ 55 \\ 50 \end{pmatrix}$$

subtract $2\cdot$(row 1) from (row 2):

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad E_{21}A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix} \qquad E_{21}\mathbf{b} = \begin{pmatrix} 19 \\ 17 \\ 50 \end{pmatrix}$$

subtract $1\cdot$(row 1) from (row 3):

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \qquad E_{31}E_{21}A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & 5 & 13 \end{bmatrix} \qquad E_{31}E_{21}\mathbf{b} = \begin{pmatrix} 19 \\ 17 \\ 31 \end{pmatrix}$$

subtract $1\cdot$(row 2) from (row 3):

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \qquad \underbrace{E_{32}E_{31}E_{21}A}_{U} = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & \mathbf{7} \end{bmatrix} \qquad \underbrace{E_{32}E_{31}E_{21}\mathbf{b}}_{\mathbf{c}} = \begin{pmatrix} 19 \\ 17 \\ 14 \end{pmatrix}$$

↑ elimination matrices

done! Now back substitution...

Row exchanges to deal with zero pivots: also undoable, solutions stay the same

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 8 & 17 \end{bmatrix} \qquad \mathbf{b} = \cdots$$

elimination in column 1:

$$E_{31}E_{21}A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{0} & 6 \\ 0 & 5 & 13 \end{bmatrix} \qquad E_{31}E_{21}\mathbf{b} = \cdots$$

pivot is **0**: exchange (row 2) and (row 3):

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad \underbrace{P_{23}E_{31}E_{21}A}_{U} = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{5} & 13 \\ 0 & 0 & 6 \end{bmatrix} \qquad \underbrace{P_{23}E_{31}E_{21}\mathbf{b}}_{\mathbf{c}} = \cdots$$

↑ permutation matrix

done!

Failure: no row exchange helps, give up!

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 3 & 17 \end{bmatrix} \qquad \mathbf{b} = \cdots$$

elimination in column 1:

$$E_{31}E_{21}A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{0} & 6 \\ 0 & 0 & 13 \end{bmatrix} \qquad E_{31}E_{21}\mathbf{b} = \cdots$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & \mathbf{0} \end{bmatrix}$$

we can also fail in the last column!

Algorithm 2:

function GAUSS ELIMINATION($A, \mathbf{b}$) ▷ $A \in \mathbb{R}^{m\times m}, \mathbf{b} \in \mathbb{R}^m$
    $U \leftarrow A, \mathbf{c} \leftarrow \mathbf{b}$
    for $j = 1, 2, \ldots, m$ **do** ▷ eliminate in column j
        if $u_{jj} = 0$ **then** ▷ zero pivot
            if there is some $k > j$ such that $u_{kj} \neq 0$ **then**
                exchange (row j) and (row k) (in both U and $\mathbf{c}$) ▷ row operation
            else ▷ give up
                return $(U, \mathbf{c},$ "failed"$)$
            end if
        end if ▷ now, $u_{jj} \neq 0$
        for $i = j+1, j+2, \ldots, m$ **do** ▷ make $u_{ij} = 0$
            $\lambda \leftarrow \frac{u_{ij}}{u_{jj}}$ ▷ we want $u_{ij} - \lambda u_{jj} = 0$
            subtract $\lambda \cdot$ (row j) from (row i) (in both U and $\mathbf{c}$) ▷ row operation
        end for
    end for ▷ now, U is upper triangular, with all diagonal elements nonzero
    return $(U, \mathbf{c},$ "succeeded"$)$
end function

Worked example with nice numbers:
`https://ti.inf.ethz.ch/ew/courses/LA25/slides/Gauss_nice.pdf`
Worked example with ugly numbers:
`https://ti.inf.ethz.ch/ew/courses/LA25/slides/Gauss_ugly.pdf`
Worked example where elimination fails:
`https://ti.inf.ethz.ch/ew/courses/LA25/slides/Gauss_failure.pdf`

Row operations, the general picture:
Lemma 3.2 (Invariance of solutions): Let A be an $m \times n$ matrix, M an invertible $m \times m$ matrix, $\mathbf{b} \in \mathbb{R}^m$. The two systems $A\mathbf{x} = \mathbf{b}$ and $MA\mathbf{x} = M\mathbf{b}$ have the same solutions $\mathbf{x}$.

Proof. For every $\mathbf{x} \in \mathbb{R}^n$,

- $A\mathbf{x} = \mathbf{b} \Rightarrow MA\mathbf{x} = M\mathbf{b}$;
- $MA\mathbf{x} = M\mathbf{b} \Rightarrow \underbrace{\overbrace{M^{-1}M}^{I}\, A\mathbf{x}}_{A\mathbf{x}} = \underbrace{\overbrace{M^{-1}M}^{I}\, \mathbf{b}}_{\mathbf{b}}$.

□

Lemma 3.3 (Invariance of the nullspace): A, M as before. Then A and MA have the same nullspace, $\mathbf{N}(A) = \mathbf{N}(MA)$.

Proof. Apply Lemma 3.2 with $\mathbf{b} = \mathbf{0}$. □

Lemma 3.4 (Invariance of linear independence): A, M as before. A has linearly independent columns if and only if MA has linearly independent columns.

Proof. Columns are linearly independent $\Leftrightarrow$ nullspace $= \{\mathbf{0}\}$ (Obs. 2.5). Statement therefore follows from Lemma 3.3. $\square$

Lemma 3.5 (Invariance of the row space): A, M as before. Then A and MA have the same row space, $\mathbf{R}(A) = \mathbf{R}(MA)$ (see exercise sessions).

However, $\mathbf{C}(A) \neq \mathbf{C}(MA)$ in general (see exercise sessions).

Lemma 3.6 (Invariance of independent column indices and rank): A, M as before. For all $j \in [n]$: column j of A is independent $\Leftrightarrow$ column j of MA is independent. So A and MA have the same number of independent columns and therefore the same rank.

Proof.

$$A = \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_m \\ | & | & & | \end{bmatrix} \qquad MA = \begin{bmatrix} | & | & & | \\ M\mathbf{v}_1 & M\mathbf{v}_2 & \cdots & M\mathbf{v}_m \\ | & | & & | \end{bmatrix}$$

$$B = \begin{bmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_{j-1} \\ | & | & & | \end{bmatrix} \qquad MB = \begin{bmatrix} | & | & & | \\ M\mathbf{v}_1 & M\mathbf{v}_2 & \cdots & M\mathbf{v}_{j-1} \\ | & | & & | \end{bmatrix}$$

$$\begin{array}{ccc} \mathbf{v}_j \text{ independent in } A & & M\mathbf{v}_j \text{ independent in } MA \\ \Updownarrow & & \Updownarrow \\ B\mathbf{x} = \mathbf{v}_j \text{ has no solution} & \overset{\text{(Lemma 3.2)}}{\Leftrightarrow} & MB\mathbf{x} = M\mathbf{v}_j \text{ has no solution} \end{array}$$

$\square$

Success and failure:
Theorem 3.7: Let $A\mathbf{x} = \mathbf{b}$ be a system of m linear equations in m variables. The following two statements are equivalent.

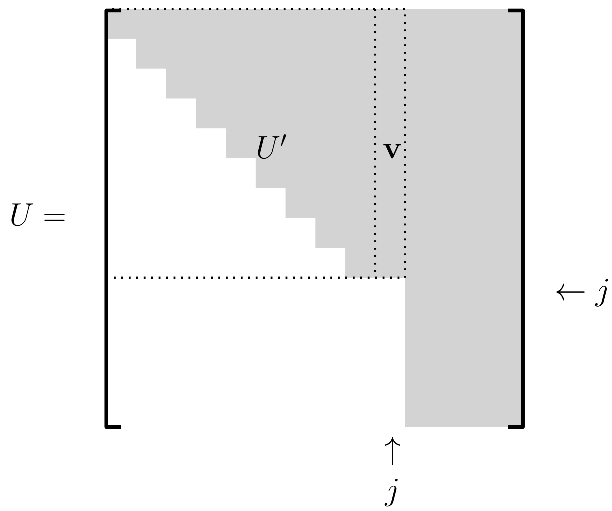
(i) Gauss elimination as in Algorithm 2 succeeds.

(ii) The columns of A are linearly independent.

We prove (i)⇒(ii) and ¬(i)⇒ ¬(ii): if *not* (i), then *not* (ii). This is the *contraposition* of (ii)⇒(i) and logically equivalent.

Proof.
(i)⇒(ii): If Gauss elimination succeeds: $A \to$ upper triangular U with all $u_{jj} \neq 0$. U has linearly independent columns by Corollary 1.23 (iii): no column is a linear combination of the previous ones. We get (ii) by Lemma 3.4 (invariance of linear independence), repeatedly applied ($A \to A' \ldots \to U$).

¬(i)⇒ ¬(ii): If Gauss elimination fails in column j, we have U with $u_{11}, u_{22}, u_{j-1,j-1} \neq 0$:



- $\mathbf{v}$ is a linear combination of the columns of U' (back substitution on $U'\mathbf{x} = \mathbf{v}$).
- Because of 0's below U' and $\mathbf{v}$: this also writes j-th column of U as a linear combination of the previous columns.
- Columns of U (and hence of original A; Lemma 3.4) are linearly dependent; this is $\neg$(ii).

$\square$

Runtime for solving $A\mathbf{x} = \mathbf{b}$: Count basic operations ($-, \cdot, /$, variable updates,...) in O-notation!
Elimination:

- At most m^2 row subtractions (one for every matrix entry below the diagonal)
- $O(m)$ basic operations per row subtraction
- Time $m^2 \cdot O(m) = O(m^3)$ in total (also covers the other operations, for example the at most m row exchanges)

Back substitution:

- Computation of m values
- $O(m)$ basic operations per value
- Time $m \cdot O(m) = O(m^2)$ in total

Together: $O(m^3) + O(m^2) = O(m^3)$.

Computing A^{-1} in time $O(m^4)$: solve m systems $A\mathbf{x} = \mathbf{e}_1, A\mathbf{x} = \mathbf{e}_2, \ldots, A\mathbf{x} = \mathbf{e}_m$.
All systems have *the same A*. *One* elimination with m right-hand sides suffices and gives time $O(m^3)$ (Algorithms 3 and 4).